

LOGISTIZATION OF MEDICAL AND PREVENTIVE INSTITUTION ACTIVITIES: STRATEGIC ANALYSIS AND OPTIMIZATION PATHS

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ABSTRACT

This article examines the complex issues related to the logistization of activities in medical and preventive institutions (MPIs). The study analyzes the dynamics of the 2015–2025 period, public health indicators, and the economic efficiency of the healthcare system. The work focuses on optimizing logistical potential through the introduction of modern models for managing patient flows, as well as material and financial resources. The article presents research findings confirming the decisive role of logistical transformation in improving the quality of medical services.

Keywords: Medical Logistization, Logistical Potential.

INTRODUCTION

In today's globalized environment, the healthcare system faces a significant challenge: how to deliver high-quality medical services with limited resources. The logistization of medical and preventive institutions represents an innovative way to address this challenge. Medical logistics is no longer viewed merely as a "supply chain" but as a strategic management tool that integrates medical, social, and economic considerations.

The relevance of this work stems from changing public morbidity patterns, an increasing prevalence of chronic diseases, and the growing need for patient-oriented, flexible logistical systems. The purpose of the study is to identify the relationship between an institution's logistical throughput and public health indicators.

A comprehensive diagnosis of the current situation is the first prerequisite for effective healthcare logistization. The analysis of the 2015–2025 period shows that although Georgia's healthcare system is undergoing transformation, significant issues remain. Particularly noteworthy is the high mortality rate among the working-age population—80% of whom are male. This suggests that existing medical logistics systems fail to ensure timely preventive interventions.

Diseases of the circulatory system (ischemic heart disease, stroke) and malignant tumors continue to dominate morbidity patterns. The share of mental disorders has reached 16%, largely influenced by social and environmental stress factors. Strengthening logistical services in this direction—such as telemedicine and outpatient monitoring—is crucial to preventing hospital overcrowding.

RESEARCH METHODOLOGY

The study employed systematic analysis, statistical observation, and comparative economic analysis. The research focused on three main directions:

1. Study of morbidity dynamics:

2. Ten-year data were analyzed across various nosological groups.

3. Resource audit:

4. The material-technical base of MPIs, the condition of medical equipment, and staff qualifications were evaluated.

5. Identification of logistical flows:

6. Patient pathways were modeled from the primary level through rehabilitation.

7. The research revealed a significant imbalance: while 70% of high-tech equipment is concentrated in Tbilisi, logistical accessibility in the regions is 40% lower. This results in the irrational migration of patients to the capital, increasing service costs and delaying timely care.

ECONOMIC ANALYSIS

Public healthcare expenditure in Georgia is approximately 6% of GDP—significantly lower than the EU average of 11%. Optimizing logistical potential can help reduce these disparities. Simulation results show that investing 1 GEL in preventive measures through well-designed screening centers saves 7 GEL in future treatment costs.

Supply Chain Management (SCM) plays a central role in logistization. Optimizing inventory management of medicines and consumables can reduce operating expenses by 15–20%, directly improving the affordability of medical services.

The study also examined international healthcare models:

- **German model:**

- Emphasizes preventive logistics. Screening coverage of 70% significantly reduces hospitalization needs.

- **French model:**

- Highlights digital logistics. The “Carte Vitale” system integrates all patient data, reducing duplicate tests by 30%.

A hybrid model adapted to Georgia’s geographical and economic conditions is recommended.

STRATEGIC DIRECTIONS AND OPTIMIZATION PATHS

Based on the research findings, several strategic directions were identified:

1. Development of an integrated information system

A unified electronic patient history available across all levels of care will eliminate information gaps and ensure seamless data flow.

2. Logistization of primary healthcare

Strengthening the role of the family doctor as a “logistical hub” will enable efficient coordination of patient flows toward specialized services.

3. Implementation of telemedicine in regions;

Remote consultations will help patients overcome territorial barriers, reduce transportation costs, and accelerate diagnostics.

Implementing international best practices—such as Germany’s screening model and France’s digital integration system—will enhance the resilience and sustainability of Georgia’s healthcare system.

The findings confirm that traditional management methods are insufficient. Logistization is essential for reducing resource waste and improving public health outcomes. Incorporating logistical principles will enable medical and preventive institutions to achieve long-term sustainability and deliver higher-quality care.

CONCLUSION

The study confirms that the logistization of medical and preventive institutions is not merely a technical adjustment but a systemic reform. Analysis of 2015–2025 data shows that conventional management approaches are exhausted. Optimizing logistical potential will:

- Improve the speed and quality of medical services
- Minimize inefficient resource expenditure
- Enhance the country's demographic situation through stronger preventive care

Ultimately, effective medical logistics is the foundation of a healthy society and a strong economy and should be a key direction for the development of Georgia's healthcare sector.

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სამკურნალო-პროფილაქტიკური დაწესებულებების საქმიანობის ლოგისტიკა: სტრატეგიული ანალიზი და ოპტიმიზაციის საკითხები

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აბსტრაქტი

განხილულია სამკურნალო-პროფილაქტიკური დაწესებულებების (სპდ) საქმიანობის ლოგისტიკის კომპლექსური საკითხები. კვლევის ფარგლებში გაანალიზებულია 2015-2025 წლების პერიოდის დინამიკა, საზოგადოებრივი ჯანმრთელობის ინდიკატორები და ჯანდაცვის სისტემის ეკონომიკური ეფექტიანობა. ნაშრომი ორიენტირებულია ლოგისტიკური პოტენციალის ოპტიმიზაციაზე, რაც გულისხმობს პაციენტთა ნაკადების, მატერიალური და ფინანსური რესურსების მართვის თანამედროვე მოდელების დანერგვას. სტატიაში წარმოდგენილია კვლევის შედეგები, რომლებიც ადასტურებენ ლოგისტიკური ტრანსფორმაციის გადამწყვეტ როლს სამედიცინო მომსახურების ხარისხის ამაღლებაში.

საკვანძო სიტყვები: სამედიცინო ლოგისტიკა, ლოგისტიკის პოტენციალი.